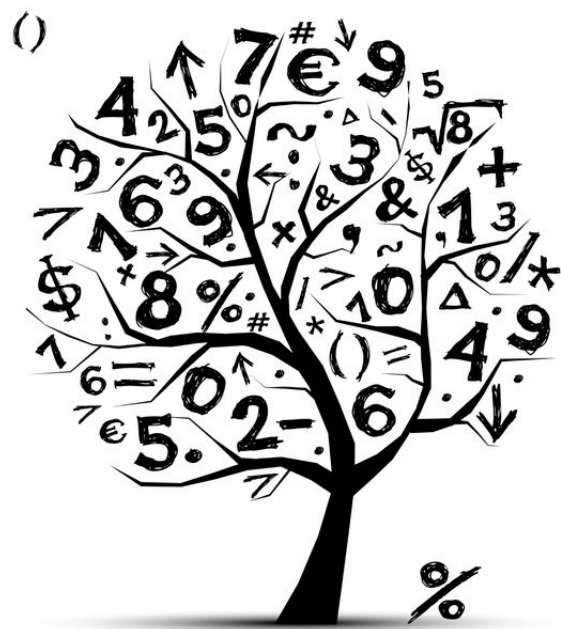


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Vectors



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1 Definition and transpose

Definition 1: Vector

A **vector** is a column of real numbers with n rows.

Given a vector $\mathbf{a} \in \mathbb{R}^n$, we write:

$$\mathbf{a} = \begin{pmatrix} [\mathbf{a}]_1 \\ [\mathbf{a}]_2 \\ \vdots \\ [\mathbf{a}]_n \end{pmatrix} \quad \text{or, for brevity,} \quad \mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}.$$

- The integer n is the **dimension** of the vector (the number of rows).
- The quantity $[\mathbf{a}]_i$ (or a_i) denotes the **i -th entry** (or **i -th component**) of $\mathbf{a}$, for $i \in \{1, 2, \dots, n\}$.

Example 1: Vector

The vector

$$\mathbf{a} = \begin{pmatrix} 2 \\ -5 \\ 3 \\ 1 \end{pmatrix} \in \mathbb{R}^4$$

is a column vector with $n = 4$ rows, where $a_1 = 2$, $a_2 = -5$, $a_3 = 3$, and $a_4 = 1$.

Definition 2: Transpose of a vector

Given a column vector $\mathbf{a} \in \mathbb{R}^n$, the **transpose** of $\mathbf{a}$, denoted by $\mathbf{a}'$, is the row vector:

$$\mathbf{a}' = (a_1 \quad a_2 \quad \dots \quad a_n) \in \mathbb{R}^{1 \times n}. \quad (1)$$

Example 2: Transpose of a vector

Given the column vector

$$\mathbf{a} = \begin{pmatrix} 2 \\ -5 \\ 3 \\ 1 \end{pmatrix} \in \mathbb{R}^4,$$

its transpose is the row vector

$$\mathbf{a}' = (2 \quad -5 \quad 3 \quad 1) \in \mathbb{R}^{1 \times 4}.$$

2 Sum of vectors and scalar multiplication

Given two column vectors

$$\underbrace{\begin{pmatrix} p_1 \\ p_2 \\ \vdots \\ p_n \end{pmatrix}}_{\mathbf{p}} \in \mathbb{R}^n \quad \text{and} \quad \underbrace{\begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{pmatrix}}_{\mathbf{w}} \in \mathbb{R}^n$$

their **sum** is the vector

$$\mathbf{p} + \mathbf{w} = \begin{pmatrix} p_1 + w_1 \\ p_2 + w_2 \\ \vdots \\ p_n + w_n \end{pmatrix} \in \mathbb{R}^n \quad (2)$$

Given a scalar $\lambda \in \mathbb{R}$ and a column vector

$$\underbrace{\begin{pmatrix} p_1 \\ p_2 \\ \vdots \\ p_n \end{pmatrix}}_{\mathbf{p}} \in \mathbb{R}^n$$

the **scalar multiplication** of $\mathbf{p}$ by λ is the vector

$$\lambda \mathbf{p} = \begin{pmatrix} \lambda p_1 \\ \lambda p_2 \\ \vdots \\ \lambda p_n \end{pmatrix} \in \mathbb{R}^n \quad (3)$$

Example 3: Sum of vectors and scalar multiplication

Consider the following two vectors and a scalar:

$$\mathbf{p} = \begin{pmatrix} 1 \\ 4 \end{pmatrix} \in \mathbb{R}^2, \quad \mathbf{w} = \begin{pmatrix} 4 \\ 1 \end{pmatrix} \in \mathbb{R}^2, \quad \lambda = 2.$$

Then:

$$\mathbf{p} + \mathbf{w} = \begin{pmatrix} 1+4 \\ 4+1 \end{pmatrix} = \begin{pmatrix} 5 \\ 5 \end{pmatrix}, \quad \lambda \mathbf{p} = 2 \begin{pmatrix} 1 \\ 4 \end{pmatrix} = \begin{pmatrix} 2 \\ 8 \end{pmatrix}.$$

2.1 Properties

The properties of vector addition and scalar multiplication are:

$$\mathbf{p} + \mathbf{w} = \mathbf{w} + \mathbf{p}, \quad \forall \mathbf{p}, \mathbf{w} \in \mathbb{R}^n \quad (4)$$

$$(\mathbf{p} + \mathbf{w}) + \mathbf{u} = \mathbf{p} + (\mathbf{w} + \mathbf{u}), \quad \forall \mathbf{p}, \mathbf{w}, \mathbf{u} \in \mathbb{R}^n \quad (5)$$

$$\lambda(\mathbf{p} + \mathbf{w}) = \lambda\mathbf{p} + \lambda\mathbf{w}, \quad \forall \lambda \in \mathbb{R}, \mathbf{p}, \mathbf{w} \in \mathbb{R}^n \quad (6)$$

$$(\lambda + \mu)\mathbf{p} = \lambda\mathbf{p} + \mu\mathbf{p}, \quad \forall \lambda, \mu \in \mathbb{R}, \mathbf{p} \in \mathbb{R}^n \quad (7)$$

$$\lambda(\mu\mathbf{p}) = (\lambda\mu)\mathbf{p}, \quad \forall \lambda, \mu \in \mathbb{R}, \mathbf{p} \in \mathbb{R}^n \quad (8)$$

$$1 \cdot \mathbf{p} = \mathbf{p}, \quad \forall \mathbf{p} \in \mathbb{R}^n. \quad (9)$$

3 Scalar product

Given two column vectors

$$\underbrace{\begin{pmatrix} p_1 \\ p_2 \\ \vdots \\ p_n \end{pmatrix}}_{\mathbf{p}} \in \mathbb{R}^n \quad \text{and} \quad \underbrace{\begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{pmatrix}}_{\mathbf{w}} \in \mathbb{R}^n$$

the expression

$$\mathbf{p}'\mathbf{w} = \sum_{j=1}^n p_j w_j \quad (10)$$

is called the **scalar product** of $\mathbf{p}$ and $\mathbf{w}$.

Example 4: Scalar product

Consider the two vectors $\begin{pmatrix} 1 \\ 4 \end{pmatrix} \in \mathbb{R}^2$ and $\begin{pmatrix} 4 \\ 1 \end{pmatrix} \in \mathbb{R}^2$. Their scalar product is:

$$(1 \quad 4) \begin{pmatrix} 4 \\ 1 \end{pmatrix} = 1 \cdot 4 + 4 \cdot 1 = 4 + 4 = 8.$$

3.1 Properties

The properties of the scalar product are:

$$\mathbf{p}' \mathbf{w} = \mathbf{w}' \mathbf{p}, \quad \forall \mathbf{p}, \mathbf{w} \in \mathbb{R}^n \quad (11)$$

$$\mathbf{p}' (\mathbf{w} + \mathbf{u}) = \mathbf{p}' \mathbf{w} + \mathbf{p}' \mathbf{u}, \quad \forall \mathbf{p}, \mathbf{w}, \mathbf{u} \in \mathbb{R}^n \quad (12)$$

$$\lambda (\mathbf{p}' \mathbf{w}) = (\lambda \mathbf{p})' \mathbf{w}, \quad \forall \lambda \in \mathbb{R}, \mathbf{p}, \mathbf{w} \in \mathbb{R}^n \quad (13)$$

From the definition we have:

$$\mathbf{p}' \mathbf{p} \geq 0, \quad \forall \mathbf{p} \in \mathbb{R}^n \quad \text{and} \quad \mathbf{p}' \mathbf{p} = 0 \iff \mathbf{p} = \mathbf{0} \quad (14)$$

4 Linear combinations and linear independence

Definition 3: Linear combination

Given k vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in \mathbb{R}^n$ and k scalars $\lambda_1, \lambda_2, \dots, \lambda_k \in \mathbb{R}$, the vector

$$\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_k \mathbf{v}_k = \sum_{i=1}^k \lambda_i \mathbf{v}_i \in \mathbb{R}^n \quad (15)$$

is called a **linear combination** of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ with **coefficients** $\lambda_1, \lambda_2, \dots, \lambda_k$.

- We denote by $\mathbf{0} \in \mathbb{R}^n$ the **zero vector**, i.e., the vector whose entries are all equal to 0.
- Choosing $\lambda_1 = \lambda_2 = \dots = \lambda_k = 0$ we always obtain $\mathbf{0}$: this is called the **trivial** linear combination.

Example 5: Linear combination

Consider the vectors

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \in \mathbb{R}^3,$$

and the coefficients $\lambda_1 = 2$ and $\lambda_2 = -3$. The corresponding linear combination is

$$2\mathbf{v}_1 - 3\mathbf{v}_2 = \begin{pmatrix} 2 \\ 4 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ -3 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}.$$

Definition 4: Linearly independent vectors

The vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in \mathbb{R}^n$ are **linearly independent** if the only linear combination equal to the zero vector is the trivial one, i.e.,

$$\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_k \mathbf{v}_k = \mathbf{0} \implies \lambda_1 = \lambda_2 = \dots = \lambda_k = 0. \quad (16)$$

Otherwise, i.e., if there exist coefficients $\lambda_1, \lambda_2, \dots, \lambda_k$, **not all zero**, such that $\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_k \mathbf{v}_k = \mathbf{0}$, the vectors are **linearly dependent**.

Example 6: Linearly independent vectors in $\mathbb{R}^2$

Consider the vectors $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} 3 \\ 1 \end{pmatrix} \in \mathbb{R}^2$. The condition $\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 = \mathbf{0}$ reads

$$\begin{cases} \lambda_1 + 3\lambda_2 = 0 \\ 2\lambda_1 + \lambda_2 = 0 \end{cases}$$

From the first equation $\lambda_1 = -3\lambda_2$; substituting in the second one we get $-6\lambda_2 + \lambda_2 = -5\lambda_2 = 0$, hence $\lambda_2 = 0$ and $\lambda_1 = 0$. Therefore, $\mathbf{v}_1$ and $\mathbf{v}_2$ are linearly independent.

Example 7: Linearly dependent vectors in $\mathbb{R}^2$

Consider the vectors

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 3 \\ 1 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 4 \\ 3 \end{pmatrix} \in \mathbb{R}^2.$$

With the coefficients $\lambda_1 = 1$, $\lambda_2 = 1$, $\lambda_3 = -1$ (not all zero) we get

$$\mathbf{v}_1 + \mathbf{v}_2 - \mathbf{v}_3 = \begin{pmatrix} 1+3-4 \\ 2+1-3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Therefore, $\mathbf{v}_1$, $\mathbf{v}_2$ and $\mathbf{v}_3$ are linearly dependent.

Example 8: Linearly independent vectors in $\mathbb{R}^3$

Consider the vectors

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \in \mathbb{R}^3.$$

The condition $\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \lambda_3 \mathbf{v}_3 = \mathbf{0}$ reads

$$\begin{cases} \lambda_1 + \lambda_3 = 0 \\ \lambda_2 + \lambda_3 = 0 \\ \lambda_1 + \lambda_2 = 0 \end{cases}$$

From the first two equations $\lambda_1 = -\lambda_3$ and $\lambda_2 = -\lambda_3$; substituting in the third one we get $-2\lambda_3 = 0$. Hence $\lambda_1 = \lambda_2 = \lambda_3 = 0$, and the three vectors are linearly independent.

Example 9: Linearly dependent vectors in $\mathbb{R}^3$

Consider the vectors

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 7 \\ 8 \\ 9 \end{pmatrix} \in \mathbb{R}^3.$$

With the coefficients $\lambda_1 = 1$, $\lambda_2 = -2$, $\lambda_3 = 1$ we get

$$\mathbf{v}_1 - 2\mathbf{v}_2 + \mathbf{v}_3 = \begin{pmatrix} 1-8+7 \\ 2-10+8 \\ 3-12+9 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

Therefore, the three vectors are linearly dependent. Equivalently, $\mathbf{v}_3 = 2\mathbf{v}_2 - \mathbf{v}_1$ is a linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$.

Observation 1: Sets containing the zero vector

Any set of vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in \mathbb{R}^n$ containing the zero vector $\mathbf{0}$ is linearly dependent.

Proof. Assume, without loss of generality, that $\mathbf{v}_1 = \mathbf{0}$. Choosing $\lambda_1 = 1$ and $\lambda_2 = \dots = \lambda_k = 0$ we get $1 \cdot \mathbf{0} + 0 \cdot \mathbf{v}_2 + \dots + 0 \cdot \mathbf{v}_k = \mathbf{0}$, which is a linear combination equal to $\mathbf{0}$ with coefficients not all zero. $\square$

Observation 2: Characterization of linear dependence

Let $k \geq 2$. The vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in \mathbb{R}^n$ are linearly dependent if and only if at least one of them is a linear combination of the others.

Proof. ($\Rightarrow$) If the vectors are linearly dependent, there exist coefficients, not all zero, such that $\sum_{i=1}^k \lambda_i \mathbf{v}_i = \mathbf{0}$. Let j be an index with $\lambda_j \neq 0$. Then

$$\mathbf{v}_j = \sum_{i \neq j} \left(-\frac{\lambda_i}{\lambda_j} \right) \mathbf{v}_i,$$

i.e., $\mathbf{v}_j$ is a linear combination of the others.

($\Leftarrow$) If $\mathbf{v}_j = \sum_{i \neq j} \mu_i \mathbf{v}_i$ for some scalars μ_i , then $\sum_{i \neq j} \mu_i \mathbf{v}_i + (-1) \mathbf{v}_j = \mathbf{0}$, which is a linear combination equal to $\mathbf{0}$ whose coefficient of $\mathbf{v}_j$ is $-1 \neq 0$. $\square$

Observation 3: More than n vectors of $\mathbb{R}^n$

Any $k > n$ vectors of $\mathbb{R}^n$ are linearly dependent.

- We state this result without proof. For instance, any three vectors of $\mathbb{R}^2$ are linearly dependent (see Example 7).

4.1 Span and basis**Definition 5: Span**

The **span** of the vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in \mathbb{R}^n$ is the set of all their linear combinations:

$$\text{span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k) = \left\{ \lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_k \mathbf{v}_k : \lambda_1, \lambda_2, \dots, \lambda_k \in \mathbb{R} \right\}. \quad (17)$$

Definition 6: Basis of $\mathbb{R}^n$

The vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in \mathbb{R}^n$ form a **basis** of $\mathbb{R}^n$ if

- they are linearly independent, and

- $\text{span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k) = \mathbb{R}^n$, i.e., every vector of $\mathbb{R}^n$ is a linear combination of them.

Observation 4: Coordinates with respect to a basis

If $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ form a basis of $\mathbb{R}^n$, then every vector $\mathbf{x} \in \mathbb{R}^n$ can be written in a **unique** way as $\mathbf{x} = \lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_k \mathbf{v}_k$. The coefficients $\lambda_1, \lambda_2, \dots, \lambda_k$ are called the **coordinates** of $\mathbf{x}$ with respect to the basis.

Proof. Such coefficients exist because the span of the basis is $\mathbb{R}^n$. If $\mathbf{x} = \sum_{i=1}^k \lambda_i \mathbf{v}_i = \sum_{i=1}^k \mu_i \mathbf{v}_i$, subtracting the two expressions we get $\sum_{i=1}^k (\lambda_i - \mu_i) \mathbf{v}_i = \mathbf{0}$. Since the vectors are linearly independent, $\lambda_i - \mu_i = 0$ for all $i \in \{1, 2, \dots, k\}$. $\square$

Definition 7: Canonical basis

For $i \in \{1, 2, \dots, n\}$, we denote by $\mathbf{e}_i \in \mathbb{R}^n$ the vector whose i -th entry is equal to 1 and all other entries are equal to 0:

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \quad \dots, \quad \mathbf{e}_n = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}.$$

The vectors $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ form the **canonical basis** of $\mathbb{R}^n$.

Observation 5: The canonical basis is a basis

The vectors $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ form a basis of $\mathbb{R}^n$, and every $\mathbf{x} \in \mathbb{R}^n$ satisfies

$$\mathbf{x} = x_1 \mathbf{e}_1 + x_2 \mathbf{e}_2 + \dots + x_n \mathbf{e}_n. \quad (18)$$

Proof. For any scalars $\lambda_1, \dots, \lambda_n$ we have $\lambda_1 \mathbf{e}_1 + \dots + \lambda_n \mathbf{e}_n = (\lambda_1 \ \lambda_2 \ \dots \ \lambda_n)'$. This vector is equal to $\mathbf{0}$ only if $\lambda_1 = \dots = \lambda_n = 0$, hence the vectors are linearly independent; choosing $\lambda_i = x_i$ we obtain (18), hence their span is $\mathbb{R}^n$. $\square$

- It can be shown that every basis of $\mathbb{R}^n$ consists of exactly n vectors.

Example 10: Canonical basis of $\mathbb{R}^3$

The vector $\mathbf{x} = (2 \ -5 \ 3)' \in \mathbb{R}^3$ can be written as

$$\mathbf{x} = 2 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} - 5 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + 3 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = 2\mathbf{e}_1 - 5\mathbf{e}_2 + 3\mathbf{e}_3.$$

Example 11: Coordinates with respect to a basis of $\mathbb{R}^2$

Consider the vectors $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

- They are linearly independent: $\lambda_1 + \lambda_2 = 0$ and $\lambda_1 - \lambda_2 = 0$ give $\lambda_1 = \lambda_2 = 0$.
- Their span is $\mathbb{R}^2$: for every $(a \ b)' \in \mathbb{R}^2$,

$$\begin{pmatrix} a \\ b \end{pmatrix} = \frac{a+b}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{a-b}{2} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Hence $\mathbf{v}_1, \mathbf{v}_2$ form a basis of $\mathbb{R}^2$. For instance, the coordinates of $\mathbf{x} = (3 \ 1)'$ are $\lambda_1 = \frac{3+1}{2} = 2$ and $\lambda_2 = \frac{3-1}{2} = 1$:

$$\begin{pmatrix} 3 \\ 1 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + 1 \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$